

The minimum signed spectral radius of $C_{20}(1, 2)$

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Abstract

We give an exact computer-assisted proof that the minimum spectral radius over all signings of $C_{20}(1, 2)$ is $2\sqrt{\cos^2(\pi/20) + \cos^2(\pi/10)}$. The minimizers are precisely the two switching classes with alternating triangle signs and negative Hamilton-cycle sign. Every other switching class has spectral radius strictly greater than $2743/1000$, which exceeds the minimum. The lower bound is certified by 240 integer Rayleigh witnesses on ten-vertex paths and a sixteen-state overlap graph; four exceptional classes are resolved by exact characteristic polynomials. An additional exact replay covers all 2^{21} switching classes. The attaining construction is due to Suvagiya. The result concerns order twenty only and does not resolve the conjecture over all even orders.

1 Statement and context

Let $G = C_{20}(1, 2)$ be the simple graph on $\mathbb{Z}/20\mathbb{Z}$ with edges $\{i, i+1\}$ and $\{i, i+2\}$. It has forty edges. A signing assigns $a_i \in \{\pm 1\}$ to $\{i, i+1\}$ and $b_i \in \{\pm 1\}$ to $\{i, i+2\}$. Its adjacency matrix A is real symmetric, with these signed entries and zero elsewhere. We use the absolute spectral radius

$$\rho(A) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } A\}.$$

All subscripts on a_i, b_i and the following invariants are taken modulo twenty. Write

$$\tau_i = a_i a_{i+1} b_i, \quad \alpha = \prod_{i=0}^{19} a_i. \quad (1)$$

Thus τ_i is the sign of the triangle on $i, i+1, i+2$, and α is the sign of the Hamilton cycle consisting of the step-one edges. A switching replaces A by DAD , where D is diagonal with entries in $\{\pm 1\}$. It preserves the spectrum and both invariants.

Theorem 1. *For every signing of $C_{20}(1, 2)$,*

$$\rho(A) \geq r := 2\sqrt{\cos^2 \frac{\pi}{20} + \cos^2 \frac{\pi}{10}}. \quad (2)$$

Equality holds if and only if $\tau_{i+1} = -\tau_i$ for every i and $\alpha = -1$. These conditions describe exactly two switching classes. Every other switching class satisfies

$$\rho(A) > \frac{2743}{1000} > r. \quad (3)$$

Suvagiya's twisted construction already attains the right-hand side of (2); see Proposition 2 of [1]. The minimization over all signings is the subject of Conjecture 3 there. The inspected version reports exhaustive numerical computations at orders 8, 10, 12, 14, 16, 18, with agreement to 10^{-9} . The contribution here is an exact lower bound at order twenty, together with its equality classification and strict gap. The attaining construction is not a contribution of this paper.

As adjacent methodological context, Zeng's finite certification of a two-eigenvalue problem at order nine [2] also concerns exact finite certificates in spectral graph theory. That problem concerns the number of distinct eigenvalues of matrices with prescribed support; it supplies no theorem used in the present minimization proof. A bounded inspection of the primary source and its linked repository on September 8, 2026 found no equivalent order-twenty certificate. This is a search assessment, not a claim of priority.

2 Switching coordinates and the local certificate

Choose $d_0 = 1$ and $d_{i+1} = d_i a_i$ for $0 \leq i < 19$. Switching by $\text{diag}(d_0, \dots, d_{19})$ makes $a_0, \dots, a_{18}$ positive. A switching preserving these nineteen signs has constant diagonal and hence changes no edge sign. Therefore the free signs

$$(a_{19}, b_0, \dots, b_{19})$$

give exactly 2^{21} switching classes, covering all 2^{40} original signings. In this gauge, $a_{19} = \alpha$ and $b_i = \tau_i a_i a_{i+1}$. Thus $(\tau_0, \dots, \tau_{19}, \alpha)$ are independent, complete switching coordinates. In particular, the two alternating triangle sequences, each with either choice of α , give four classes.

For an integer $0 \leq w < 256$, let $\text{bit}_j(w)$ be its j th bit, with the least significant bit numbered zero. Define the symmetric 10×10 matrix $B(w)$ by

$$B(w)_{j,j+1} = 1 \ (0 \leq j < 9), \quad B(w)_{j,j+2} = (-1)^{\text{bit}_j(w)} \ (0 \leq j < 8), \quad (4)$$

with all other upper-triangular entries zero and zero diagonal. For $v = (v_0, \dots, v_9) \in \mathbb{Z}^{10} \setminus \{0\}$, put

$$D(v) = \sum_{j=0}^9 v_j^2, \quad N(w, v) = \left| 2 \sum_{j=0}^8 v_j v_{j+1} + 2 \sum_{j=0}^7 (-1)^{\text{bit}_j(w)} v_j v_{j+2} \right|. \quad (5)$$

Lemma 2. *If a consecutive triangle-sign window has word w , and an integer vector $v \neq 0$ satisfies*

$$\Delta(w, v) := 1000N(w, v) - 2743D(v) > 0, \quad (6)$$

then $\rho(A) > 2743/1000$.

Proof. The ten cyclically consecutive vertices $s, \dots, s+9$ induce exactly the square of a path. Indeed, for a pair at path distance at most nine, the complementary cyclic distance is at least eleven and creates no additional step-one or step-two edge. Switching successively along this path makes its step-one signs positive and its step-two signs equal to $\tau_s, \dots, \tau_{s+7}$. Its principal matrix becomes $B(w)$, where

$$w = \sum_{j=0}^7 \mathbf{1}_{\{\tau_{s+j}=-1\}} 2^j.$$

Switch v back and extend it by zero to all twenty vertices. The resulting vector V has $|V^T AV| = N(w, v)$ and $V^T V = D(v) > 0$. The spectral theorem gives $|V^T AV|/(V^T V) \leq \rho(A)$, and (6) proves the claim. This argument controls both spectral ends. $\square$

Lemma 3 (Finite integer certificate). *For every $w \in \{0, \dots, 255\} \setminus S$, where*

$$S = \{43, 45, 75, 85, 86, 90, 105, 106, 149, 150, 165, 169, 170, 180, 210, 212\}, \quad (7)$$

the accompanying certificate table contains a nonzero integer vector satisfying (6).

Exact finite verification. The file `supplement/local_certificates.json` contains 240 rows, one for each word outside S . Each row gives its word, ten integer coordinates, N and D . Substitution in (5) verifies the stored sums and (6). The minimum integer value of Δ among these rows is 3,921,348. For example, the row $w = 0$ is

$$v = (-154, -236, -325, -381, -412, -412, -381, -325, -236, -154),$$

with $N = 3,638,204$ and $D = 999,884$. The verification checks that all word keys are distinct and that their complement is exactly (7), in addition to all 240 inequalities. It uses integer arithmetic only. The table is part of the computational certificate; its accompanying verifier is described in Section 4. No spectral-radius assertion about the sixteen retained words is needed. $\square$

Suppose every eight-term window of the cyclic triangle sequence belongs to S . Successive words w, w' must satisfy

$$\left\lfloor \frac{w}{2} \right\rfloor = w' \bmod 128. \quad (8)$$

Checking (8) on S gives precisely four disjoint directed components:

$$\begin{aligned} 85 &\longrightarrow 170 \longrightarrow 85, \\ 45 &\longrightarrow 150 \longrightarrow 75 \longrightarrow 165 \longrightarrow 210 \longrightarrow 105 \longrightarrow 180 \longrightarrow 90 \longrightarrow 45, \\ 86 &\longrightarrow 43 \longrightarrow 149, \\ 169 &\longrightarrow 212 \longrightarrow 106. \end{aligned}$$

There are no other edges; 149 and 106 have no successors. The twenty windows of a cyclic sequence form a closed walk of length twenty. A terminating path cannot occur in such a walk. A walk on the directed eight-cycle cannot close after twenty steps, since $8 \nmid 20$. Only the two-cycle remains, and its words 85 and 170 are alternating. Consequently every nonalternating triangle sequence contains a word outside S , so Lemmas 2 and 3 give

$$\rho(A) > 2743/1000 \quad \text{whenever the triangle signs do not alternate.} \quad (9)$$

3 The four alternating classes

Use the path gauge above and encode its free signs by an integer mask m :

$$a_{19} = (-1)^{\text{bit}_0(m)}, \quad b_i = (-1)^{\text{bit}_{i+1}(m)} \quad (0 \leq i < 20). \quad (10)$$

The four alternating classes have masks 699050, 873813, 1223339 and 1398100. Their exact characteristic polynomials are

$$\chi_-(x) = (x^2 - 2)^2(x^8 - 18x^6 + 114x^4 - 302x^2 + 281)^2, \quad (11)$$

$$\chi_+(x) = (x^2 - 4)(x^2 - 8)(x^2 - 3)^4(x^4 - 8x^2 + 11)^2. \quad (12)$$

Here χ_- applies to masks 873813, 1223339, with $\alpha = -1$, and χ_+ to masks 699050, 1398100, with $\alpha = +1$. These identities are verified directly from the forty signed edges. For precision, start with $H_0 = I$ and for $k = 1, \dots, 20$ form

$$C_k = AH_{k-1}, \quad c_k = -\frac{\text{tr } C_k}{k}, \quad H_k = C_k + c_k I.$$

The Faddeev–LeVerrier identities give $\det(xI - A) = x^{20} + c_1 x^{19} + \dots + c_{20}$. The verifier checks each integer division, $H_{20} = 0$, and the complete coefficient vectors of (11)–(12). A separate exact check evaluated the determinants by fraction-free elimination at each integer $x = -10, \dots, 10$ for all four matrices. Agreement at these twenty-one points proves the identities for polynomials of degree at most twenty.

For $\alpha = +1$, the squared eigenvalues in (12) are 4, 8, 3, $4 - \sqrt{5}$, $4 + \sqrt{5}$. They lie in $[0, 8]$, and 8 occurs; hence $\rho(A) = \sqrt{8}$. For $\alpha = -1$, set

$$\begin{aligned} q(y) &= y^4 - 18y^3 + 114y^2 - 302y + 281, \\ t &= \frac{9 + \sqrt{5} + \sqrt{10 + 2\sqrt{5}}}{2}. \end{aligned} \tag{13}$$

The double-angle formula gives

$$r^2 = 4 + 2 \cos(\pi/10) + 2 \cos(\pi/5) = t.$$

To verify $q(t) = 0$ algebraically, put $u = \sqrt{5}$, $v = \sqrt{10 + 2u}$ and $z = u + v = 2t - 9$. Then $z^2 - 5 = 2u(z + 1)$, so

$$z^4 - 30z^2 - 40z + 5 = 0.$$

Substituting $z = 2t - 9$ gives $16q(t) = 0$.

The following values isolate all four roots of q :

$$\begin{aligned} q(2) &= 5, & q(3) &= -4, & q(4) &= 1, & q(5) &= -4, \\ q(188/25) &= -3439/390625, & q(7521/1000) &= 51117826481/10^{12}. \end{aligned}$$

Thus there is a root in each of

$$(2, 3), \quad (3, 4), \quad (4, 5), \quad (188/25, 7521/1000).$$

They are distinct; degree four shows these are all roots, and the last interval contains the largest. Exact squaring of positive rationals gives

$$\begin{aligned} \frac{2236067}{10^6} &< \sqrt{5} < \frac{2236068}{10^6}, \\ \frac{3804225}{10^6} &< \sqrt{10 + 2\sqrt{5}} < \frac{3804227}{10^6}. \end{aligned}$$

These bounds imply $188/25 < t < 7521/1000$, identifying t as the largest root of q . By (11), the squared eigenvalues are 2 and the roots of q , so the spectral radius is $\sqrt{t} = r$. Finally,

$$t < \frac{7521}{1000} < \left(\frac{2743}{1000}\right)^2 < 8. \tag{14}$$

The four alternating classes therefore have radii $r, r, \sqrt{8}, \sqrt{8}$. Together with (9), this proves all assertions of Theorem 1.

4 Reproducibility and scope

The supplemental directory contains the complete integer table, a redundant full-class coverage file, and a verifier requiring only Python’s standard library. From the manuscript directory, run

```
python3 supplement/verify.py
```

The verifier checks the local integer sums, the sixteen-state graph, the four characteristic polynomials, and rational root comparisons. For root ordering it uses the exact Sturm sequence

$$q, \quad q', \quad \frac{15y^2 - 120y + 235}{4}, \quad \frac{32y - 128}{3}, \quad \frac{5}{4},$$

whose variation counts at 0, 188/25 and 7521/1000 are 4, 1, 0. This gives the same root ordering as the elementary sign-change argument above. No numerical tolerance or numerical eigenvalue calculation is used in verification.

The file `switching_cover.bin` has exactly 2^{21} bytes. Position m corresponds to the mask in (10). Its byte is a starting vertex $s \in \{0, \dots, 19\}$ selecting a certified window, or 255 for one of the four alternating exceptions. For every ordinary entry, the verifier reconstructs the original signs, obtains the local integer vector, and lifts it to the original twenty coordinates by

$$V_{s+j} = d_j v_j \quad (0 \leq j < 10), \quad d_0 = 1, \quad d_{j+1} = d_j a_{s+j},$$

setting all other coordinates to zero. It computes the full signed-edge quadratic form

$$V^T A V = 2 \sum_{i=0}^{19} (a_i V_i V_{i+1} + b_i V_i V_{i+2})$$

and checks the strict integer inequality again. This checks all cyclic boundary windows, as well as the mask convention and local lifting.

Exact verification component	Count
Local integer witnesses	240
Retained overlap states	16
Full-class strict Rayleigh inequalities	2,097,148
Alternating polynomial exceptions	4
Total switching classes covered	2,097,152
Uncovered classes	0

The full replay is redundant for coverage: the local inequalities and the overlap graph already reduce the proof to the four exceptions. A separate verifier independently reconstructed the local matrices, formed the twentieth power of the overlap adjacency matrix, replayed every switching class, and checked the exceptional determinants as described above. It reproduced the counts and inequalities in the table.

The eight-cycle in the overlap graph is essential to the scope: this argument does not force alternation at lengths divisible by eight. No minimization theorem at another order is asserted, and the conjecture over all even orders is not resolved here. The result is an exact finite computer-assisted proof, without proof-assistant formalization.

Computational and AI assistance. AI assistance was used in proof development, code preparation, mathematical review, and manuscript preparation. Floating-point eigensolvers and rounding were used only to discover integer witness vectors. The reported conclusions use exact integer and rational verification. The separate mathematical review used a fresh model context and a separately written verifier; it is not external peer review. The packaged replay adapts the original verifier’s file handling and output without changing its mathematical checks.

References

- [1] V. Suvagiya, *Signed circulants at the Ramanujan bound*, arXiv:2607.18334v1 (2026). <https://arxiv.org/abs/2607.18334v1>.
- [2] Z. Zeng, *Exact Finite Certification of The Sparse-Complement Two-Eigenvalue Conjecture at Order Nine*, SSRN (2026). <https://doi.org/10.2139/ssrn.7379578>.